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ORIGINAL ARTICLE

The effect of dietary counselling on food intakes in pregnant women at risk for gestational diabetes: a secondary analysis of a randomised controlled trial RADIEL

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BACKGROUND/OBJECTIVES: The prevalence of gestational diabetes mellitus (GDM) is increasing worldwide. GDM may be prevented by improving the diets of pregnant women. The objective of this study was to evaluate the effect of dietary counselling on the diets of pregnant women at GDM risk.

SUBJECTS/METHODS: This study was a secondary analysis of a randomised controlled trial the Finnish gestational diabetes prevention study (RADIEL) in which pre-pregnant and pregnant women with previous GDM or BMI $\geqslant 30\ kg/m^2$ were allocated into two groups, namely the control and the intervention groups. The control group received standard antenatal dietary counselling according to the Finnish Nutrition Recommendations. The intervention group participated in one individual dietary counselling session and one group dietary counselling session in addition to the standard counselling. This study included women who were recruited during pregnancy. To assess changes in food intake, food-intake questionnaires were collected during the first and the second trimester of pregnancy. Bootstrap type analysis of covariance was used, and 242 participants were included in the final analysis to study changes in food intake.

RESULTS: The intakes of low-fat cheese (baseline adjusted mean 0.09 times/day; 95% confidence interval (CI) 0.07, 0.24; $P = 0.040$) and fish (baseline adjusted mean 0.28 times per week; 95% CI 0.08, 0.49; $P = 0.011$) showed a significant increase in the intervention group compared with the control group.

CONCLUSIONS: This study showed that dietary counselling in early pregnancy can lead to modest dietary improvements in pregnant women at GDM risk.

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INTRODUCTION

Gestational diabetes mellitus (GDM) is defined as glucose intolerance with its first detection during pregnancy.[1] Factors associated with GDM risk are pre-pregnancy overweight, previous GDM and a family history of diabetes.[2] GDM raises the risk for macrosomia, type 2 diabetes (T2D) in the mother and the risk of future obesity and diabetes in the offspring. The prevalence of GDM is increasing worldwide,[3] ranging between 1 and 18% of all pregnancies depending on the study population and the diagnostic criteria.[2,4]

A recent meta-analysis on the effectiveness of interventions in preventing GDM showed a 5% reduction in the risk of GDM among women receiving dietary counselling, when compared with women receiving usual care.[5] However, individual studies measuring the effectiveness of interventions on GDM risk have been inconclusive.[5,6]

Pregnancy may be an optimal period for interventions because pregnant women are often motivated to make lifestyle changes to benefit their unborn child.[7] Some dietary intervention studies have shown improvements in dietary intakes among pregnant women, for example, an increase in the intakes of polyunsaturated fat, dietary fibre, vegetables, fruits, whole grains and fish and a decrease in the intakes of saturated fat, sugary food, refined grain, high-fat meat and fast food.[7–12] However, the evidence for the effect of dietary counselling during pregnancy is limited. Therefore, the aim of this study was to assess whether the dietary counselling for pregnant women in the RADIEL study had positive effects on the dietary intakes of pregnant women at risk for GDM.

MATERIALS AND METHODS

Design

This study was a secondary analysis of the Finnish gestational diabetes prevention study (RADIEL), which is a randomised controlled lifestyle intervention study. The main objective of the RADIEL study was to examine

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the efficacy and cost-effectiveness of a combined diet cum physical activity intervention in preventing excessive gestational weight gain, GDM and later T2D and in reducing risk factors for cardiovascular disease. The study took place in four maternity hospitals in Finland: Helsinki University Central Hospital (HUCH); Kätilöopisto Maternity Hospital; Jorvi Hospital and the South Karelia Central Hospital (SKCH). The study was conducted between 2008 and 2014 according to the guidelines laid down in the Declaration of Helsinki, and all procedures were approved by the ethics committees of HUCH and SKCH (NCT01698385; clinicaltrials.gov). The study was originally registered before the recruitment started in 2008. However, when the RADIEL follow-up trial protocol was being added to the database, there was a technical error on the ClinicalTrials.gov website, and the registration date was accidentally changed to 2012.

Subjects

Pre-pregnant and pregnant women at risk for GDM, that is, with a history of GDM or self-reported pre-pregnancy BMI $\geqslant 30$ kg/m^2, were recruited for the RADIEL study. This study included only those women who were pregnant at the time of recruitment. Women were mainly recruited at the first-trimester ultrasound screening visit. Personal invitation letters and public notices in newspapers, social media and antenatal clinics were also used in the recruitment process. All women who participated in the study did so voluntarily and had the option to discontinue at any time. The participants were randomly assigned to an intervention or a control group after having given written informed consent. In the randomisation, randomly permuted blocks were used and stratified according to the risk factors including BMI $\geqslant 30$ kg/m^2 and a history of GDM.

The inclusion criteria were age of $\geqslant 18$ years, pregnant at < 20 weeks of gestation and a history of GDM or pre-pregnancy BMI $\geqslant 30$ kg/m^2. The criteria for exclusion were diabetes or GDM diagnosed at study inclusion, use of medication that influenced glucose metabolism, multiple pregnancy, physical disability, current substance abuse, severe psychiatric disorder or significant communication difficulties.

Interventions

The dietary intervention in this study consisted of one individual counselling session led by a trained study nurse and one 2-h group counselling session led by a dietitian. The dietary counselling was based on the contemporary Finnish Nutrition Recommendations (2005).[13] In addition to the standard visits at the antenatal clinics, the participants visited a study nurse every 3 months during the pregnancy. The first study visit was conducted during the first trimester of pregnancy (6–18 weeks of gestation), during which the study nurse gave individual counselling on gestational weight gain, physical activity and healthy diets in order to prevent GDM. She also enquired about habitual dietary intake and discussed necessary changes to the diet. After the first study visit, the participants joined a 2-h group session with six to eight participants. As the counselling led by the dietitian was delivered in groups, weeks of gestation varied between the participants. In the group session, the dietitian and the participants discussed gestational diabetes, gestational weight gain, exercise practice, dietary glycemic index and healthy diets. The dietitian also gave detailed examples of daily meals that may help the participants reach their dietary goals. The goals were to increase the intakes of vegetables, fruits, berries, vegetable fats, whole grain, fish, low-fat meat and low-fat dairy and to reduce the intake of sugar-rich foods. The intervention group also received individualised physical activity counselling at each study visit.

The study visits of the control group were similar to the standard antenatal visits that are carried by a public health nurse in Finland. The study nurse gave general information and leaflets about healthy diets and recommended physical activity, made anthropometric measurements and took blood and urine samples. The recommendations for gestational weight gain were based on the report of the Institute of Medicine.[14]

Methods

To assess food intakes, a 48-item food-intake questionnaire was developed specifically for this study. The participants completed the questionnaire and returned it to the study nurse at the first study visit (first trimester) and the second study visit (second trimester). In the questionnaire, the participants were asked to report their current frequency of consumption of various food groups, the choices ranging from 'rarely or never' to '4 or more times/day'. The food groups studied were low-fat milk ($\leqslant$1% fat), low-fat cheese ($\leqslant$17% fat), whole grain cereals (rye bread, porridge, and wholegrain rice and pasta), fruits and berries, vegetables and legumes, fish, animal protein (meat, poultry, egg and organ meats), sugar-sweetened beverages (sugary juice and soft drinks), snacks (pastries, candy, chocolate, chips and ice cream) and fast food (hamburger and pizza). To assess the quality of dietary fats, there were separate questions regarding the type of spreadable and cooking fats normally used. The responses were translated into two fat scores. Spreadable fat was scored as follows: 2 points for regular or low-fat margarine, 1 point for sterol margarine and 0 points for butter, butter-vegetable oil mixture or no spreadable fat. The scores for cooking fat were as follows: 1 point for oil, regular margarine, liquid margarine or no cooking fat and 0 points for butter, butter-vegetable oil mixture or hard margarine. In addition, the participants filled in a 3-day dietary record at the study inclusion. The dietary records were analysed and nutrient intakes calculated by two nutritionists using the food calculation software AivoDiet (Aivo Ltd, Turku, Finland), which in turn uses the Fineli Finnish Food Composition Database (National Institute for Health and Welfare, Nutrition Unit, Helsinki, Finland).

Subject inclusion

Of those 293 pregnant women who were randomised into an intervention or a control group, 144 women in the intervention group and 125 women in the control group were followed at least until delivery (Figure 1). As GDM is usually diagnosed at the second trimester of pregnancy (normally 14–26 weeks of gestation), and the diagnosis may affect food intake, the last food-intake measurement of this study was conducted in the second trimester when women were at 20–29 weeks of gestation. We included in the analysis only those participants who completed their second food-intake questionnaires before a possible GDM diagnosis. The diagnosis was defined by one or more pathological glucose values in a 75 g 2-h oral glucose tolerance test with the following diagnostic thresholds: fasting plasma glucose $\geqslant 5.3$ mmol/l, 1-h value $\geqslant 10.0$ mmol/l and 2-h value $\geqslant 8.6$ mmol/l.[15] In the intervention group, 11 participants did not return their second-trimester food-intake questionnaire, and 1 participant returned the questionnaire at the second trimester only after the diagnostic oral glucose tolerance test. In the control group, three participants did not return their first-trimester questionnaire, and seven participants did not return their second-trimester questionnaire. In addition, five participants returned their questionnaire at the second trimester after the diagnostic oral glucose tolerance test. In the end, 132 women from the intervention group and 110 women from the control group were included in the final analysis.

Outcome variables

The variables analysed were based on the food-intake questionnaires and included the daily intakes of low-fat milk, low-fat cheese, whole grain cereals, fruits and berries, vegetables and legumes and animal protein. They also included the weekly intakes of fish, snacks, sugar-sweetened beverages and fast food, as well as the intakes of spreadable and cooking fats. The changes in the consumption of the food groups between the first and second trimesters of pregnancy were stated in terms of times per day or week and for the dietary fat as changes in the fat scores.

Statistical analysis

All data are presented as means (s.d.) for continuous variables and as frequencies *n* (%) for categorical variables. Student's *t*-test and Pearson's chi-square were used for baseline analyses. Bootstrap type analysis of covariance was used to analyse changes in food intake from the first to the second trimester of pregnancy. The level of significance was set at $P < 0.05$. All analyses were performed using the Statistical Package for the Social Sciences (SPSS for Windows, release 22, IBM Corp., Armonk, NY, USA), except for the changes in the food intake where Stata 13.1, StataCorp LP (College Station, TX, USA) was used.

RESULTS

Baseline characteristics

Baseline characteristics of the 242 pregnant women included in the study are shown in Table 1. The mean age was 32.5 years and mean pre-pregnancy BMI 31.7 kg/m^2, with 84% being overweight or obese (BMI $\geqslant 25$ kg/m^2). Baseline characteristics were similar in the intervention and control groups. Baseline nutrient intakes of both groups deviated from the contemporary Finnish Nutrition

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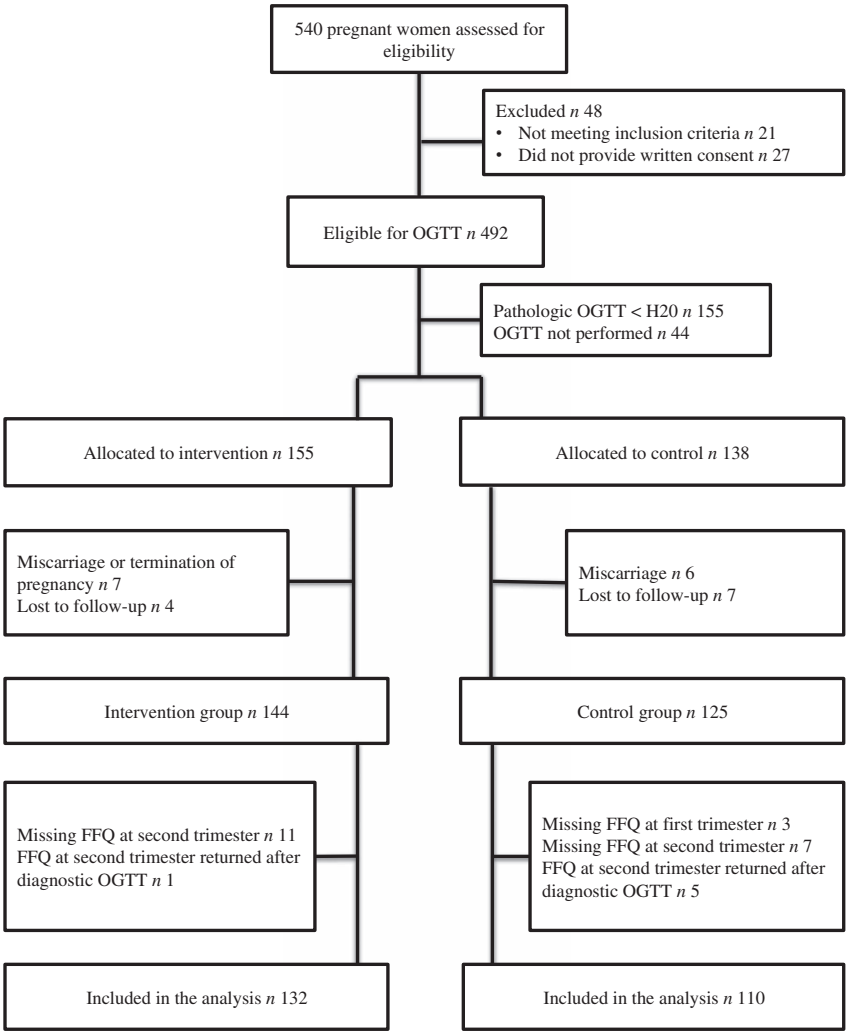

Figure 1. Flowchart of randomisation into two groups and follow-up from the first to the second trimester of pregnancy. FFQ, food frequency questionnaire; OGTT, a 75 g 2-h oral glucose tolerance test.

Recommendations (2005);[13] the mean intakes of carbohydrate and dietary fibre were below the minimum recommended levels of 50% of energy and 3 g/MJ, respectively, whereas the mean intake of saturated fat was above the recommended upper limit of 10% of energy.

Changes in food intake

The analysis of the change in food intakes between the first and second trimesters of pregnancy included 242 eligible participants. Baseline food intakes were similar between the intervention and the control groups ($P > 0.05$ for all). Dietary changes and between-group differences in food intake are shown in Table 2. The intakes of low-fat cheese ($P = 0.040$) and fish ($P = 0.011$) showed a significant increase in the intervention group compared with the control group. In addition, the intake of sugar-sweetened beverages decreased slightly in the intervention group. However, when compared with the control group, the difference was not statistically significant.

Drop out analyses

Compared with those who completed the study, the participants who dropped out ($n = 27$) had lower intakes of protein (16.6 E% compared with 17.8 E%, $P = 0.009$), folic acid (343.2 μg compared with 517.3 μg/day, $P = 0.026$) and vitamin D (8.9 μg compared with 13.8 μg/day, $P = 0.003$) at baseline. They used dietary supplements less frequently (55% compared with 78%) and had a higher intake of animal protein (10.6 times per day compared with 8.2 times per day) at baseline. Other baseline characteristics were similar between the groups.

DISCUSSION

In this study, the dietary counselling in early pregnancy, which included one individual session and one group session, had a modest effect on the diet of pregnant women at increased GDM risk. In the intervention group, there were changes in the intakes of low-fat cheese and fish, whereas the control group showed no changes in food intake.

Low-fat cheese is an excellent source of protein and calcium, brings variety to the diet and may represent reduced fat content of the diet in general. Fish is also recommended during pregnancy as long as the intake is within the guidelines pertaining to certain local species. The finding of the increased low-fat cheese intake in the intervention group has been observed in another Finnish study among women at increased GDM risk.[9] However, the increased fish consumption in the intervention group compared with the control group is contradictory to earlier studies in Finland, which showed that dietary counselling had no effect on fish intake.[9,11]

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Table 1. Baseline characteristics of the participants at inclusion[a]

	Control group[b] *(n = 110)*	*Intervention group*[c] *(n = 132)*	P-*value*[d]
Age, years	32.6 (4.6)	32.4 (4.8)	0.732
Pre-pregnancy weight, kg	88.7 (16.7)	87.0 (18.0)	0.468
Pre-pregnancy BMI, kg/m^2	32.0 (5.6)	31.3 (5.8)	0.340
Previous GDM, n *(%)*			0.609
No	29 (26)	31 (24)	
Yes	31 (28)	45 (34)	
Nullipara	50 (46)	56 (42)	
Parent's history of diabetes[e], n *(%)*			0.937
No	84 (78)	99 (77)	
Yes	24 (22)	29 (23)	
Highest education, n *(%)*			0.226
Basic or secondary education	39 (36)	58 (44)	
Polytechnic education	38 (35)	33 (25)	
Academic education	32 (29)	40 (31)	
Current smoker, *n* (%)	3 (3)	5 (4)	0.366
Dietary supplement users, *n* (%)	77 (77)	96 (80)	0.675
Nutrient intake[f]			
Energy, kcal/day	1916.6 (447.3)	1885.5 (446.5)	0.608
Protein, E%	18.0 (3.4)	17.7 (3.1)	0.480
Carbohydrates, E%	45.1 (5.7)	45.8 (6.2)	0.425
Total fat, E%	33.4 (5.9)	33.1 (6.1)	0.693
Saturated fat, E%	12.3 (3.0)	12.1 (3.1)	0.733
Sugar, E%	9.0 (4.1)	9.0 (3.7)	0.884
Fibre, g/mJ	2.9 (0.9)	2.9 (0.8)	0.926
Vitamin D, μg/day	13.7 (6.9)	13.9 (7.9)	0.862
Folic acid, μg/day	508.8 (238.2)	524.3 (268.8)	0.653
Calcium, mg/day	1311.3 (354.1)	1216.2 (431.1)	0.079

Abbreviations: BMI, body mass index; E%, % of total energy; GDM, gestational diabetes mellitus. [a]Continuous variables are presented as mean (s.d.) and categorical variables as *n* and percentages. [b]Number of missing values: highest education ($n = 1$), parent's history of diabetes ($n = 2$). [c]Number of missing values: highest education ($n = 1$), parent's history of diabetes ($n = 4$). [d]*P*-values generated by Student's *t*-test or Pearson's chi-square. [e]Type I or type II diabetes of one or both parents. [f]Data were available for 100 participants in the control group and 121 participants in the intervention group. Vitamins and minerals are presented as total intake.

Table 2. Changes in food intake from the first to the second trimester of pregnancy in the control ($n = 110$) and the intervention ($n = 132$) groups

Food component	*Baseline*[a]	*Mean change (95% CI)*	P-*value*[b]
Low-fat milk, times/day			0.726
Control	1.26 (1.11)	0.14 (−0.02, 0.29)	
Intervention	1.27 (1.30)	0.10 (−0.07, 0.28)	
Low-fat cheese, times/day			0.040
Control	0.93 (1.02)	−0.14 (−0.30, 0.01)	
Intervention	0.84 (0.97)	0.09 (0.07, 0.24)	
Whole-grain cereals, times/day			0.182
Control	2.14 (1.29)	−0.13 (−0.33, 0.07)	
Intervention	2.10 (1.41)	0.08 (−0.17, 0.33)	
Fruits and berries, times/day			0.865
Control	1.40 (1.03)	−0.03 (−0.20, 0.14)	
Intervention	1.45 (1.08)	−0.03 (−0.22, 0.15)	
Vegetables and legumes, times/day			0.419
Control	1.37 (1.11)	−0.02 (−0.21, 0.17)	
Intervention	1.49 (1.04)	−0.01 (−0.20, 0.18)	
Animal protein,[c] *times/day*			0.658
Control	1.17 (0.75)	−0.18 (−0.30, −0.06)	
Intervention	1.18 (0.89)	−0.16 (−0.30, −0.02)	
Fish, times/week			0.011
Control	1.35 (1.50)	0.06 (−0.21, 0.34)	
Intervention	1.55 (1.36)	0.28 (0.08, 0.49)	
Snacks,[d] *times/week*			0.112
Control	6.62 (6.15)	0.94 (−0.03, 1.90)	
Intervention	6.37 (4.39)	0.21 (−0.57, 0.99)	
Sugar-sweetened beverages,[e] *times/week*			0.750
Control	2.50 (5.71)	−0.95 (−1.95, 0.05)	
Intervention	2.23 (3.91)	−0.64 (−1.18, −0.10)	
Fast food,[f] *times/week*			0.731
Control	1.07 (1.42)	−0.21 (−0.41, −0.01)	
Intervention	0.85 (1.25)	−0.04 (−0.22, 0.13)	
Spread fat, score[g]			0.103
Control	1.15 (0.98)	0.15 (0.02, 0.30)	
Intervention	1.33 (0.92)	0.22 (0.11, 0.33)	
Cooking fat, score[h]			0.937
Control	0.83 (0.38)	0.05 (−0.00, 0.09)	
Intervention	0.88 (0.33)	0.02 (−0.03, 0.07)	

Abbreviation: CI, confidence interval. [a]Values given are mean (s.d.). [b]*P*-values generated by bootstrap type analysis of covariance. Adjusted for baseline. [c]Includes meat, poultry, egg and organ meats. [d]Includes pastries, candy, chocolate, chips and ice cream. [e]Includes sugared juice and soft drinks. [f]Includes hamburger and pizza. [g]Regular or low-fat margarine 2 points; sterol margarine 1 point; butter, butter-vegetable oil mixture or no spread fat 0 points. [h]Cooking oil, regular margarine, bottled margarine or no cooking fat 1 point; butter, butter-vegetable oil mixture or hard margarine 0 points.

Most studies analysing the effect of dietary counselling on diets during pregnancy suggest an improvement in dietary intake post counselling.[7–12,16–19] However, the studies differ in terms of the characteristics of the participants, the dietary measurement and counselling methods and the follow-up time and are therefore not fully comparable.

As previously reported,[20] the RADIEL study showed a significant reduction in the risk for GDM in women of this study. The reduced risk may partly be attributable to the dietary improvements reported here. However, the evidence for an association between intakes of dairy products and fish and risk for GDM is limited. A study in Massachusetts, USA, reported that the substitution of fish for red meat in pre-pregnancy was associated with a lower GDM risk, whereas the substitution of dairy products for red meat showed no association.[21] As GDM and T2D share similar risk factors, dietary factors associated with T2D may also be associated with GDM. A recent meta-analysis showed an inverse association between the intakes of low-fat dairy products and cheese and T2D risk.[22] Another recent meta-analysis showed geographical differences in the associations between fish intake and T2D risk;[23] studies in the USA showed a positive association, European studies showed no association, whereas Asian studies showed an inverse association.

The low-intensity counselling received by the intervention group in this study consisted of one individual and one group session in addition to the normal antenatal clinic visits. Although the counselling modestly increased the intake of foods associated with healthy diets, it had no effect on the intakes of energy-dense and high-saturated fatty foods such as snacks and fast food. The consumption of sugar-sweetened beverages slightly decreased in the intervention group, but the difference was not statistically significant between the groups. As high intakes of energy and saturated fat have been associated with increased GDM risk,[24] decreasing the intake of high-energy and fatty foods may have been relevant in preventing GDM.

The strengths of this study include the randomised and controlled design and the selection of the participants with

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previous GDM or pre-pregnancy BMI $\geqslant 30$ kg/m^2, which may have increased the efficacy of the counselling. By including only those participants who completed the food-intake questionnaire before the diagnostic oral glucose tolerance test and by following the participants only until the second trimester of pregnancy, we eliminated the possible effect GDM diagnosis may have on food intakes.

This study also had some limitations that may have affected the results. More intensive dietary counselling and longer follow-up may have resulted in more dietary changes. In order to make it more realistically replicable within the health-care framework in Finland, the individual counselling was provided by a study nurse. It may have been more effective if it had been provided by a dietitian instead. Moreover, all of the participants were at GDM risk and therefore received lifestyle counselling at the antenatal clinics. Thus, the control group did receive some degree of counselling as well, which may have attenuated the difference in the dietary changes between the groups. Another important factor that may have caused changes in food intake, and may therefore have affected the results, is the reduction in nausea and vomiting at the end of the first trimester of pregnancy. On the other hand, the Hawthorne effect may have occurred. This is a psychological phenomenon where participating in an experiment may affect individuals' behaviour.[25] Therefore, the participants in both groups, and especially in the intervention group, may have reported healthier food intakes than normally, making it difficult to separate the effect of the intervention from the effect of participating in the study.

Another limitation was that dietary records were not collected at the second trimester. This would have enabled the assessment of possible changes in nutrient intakes as well. According to the baseline energy intake in both groups (about 1900 kcal/day based on baseline food records; data not shown) and the prevalence of overweight and obesity (85%), under-reporting was likely. This is a common phenomenon in dietary surveys and especially among overweight and obese individuals.[26–28]

In addition, the short food-intake questionnaire may not have been comprehensive enough to capture all relevant food items, their intakes and the changes in intakes. The consumption of certain possibly interesting food groups, such as processed meat or high-fat versus low-fat meat, was not assessed because they were not included in the original version of the questionnaire. Finally, the comparison of the baseline nutrient and food intakes showed that women who dropped out of the study complied less with dietary guidelines than the women who stayed in the study. Therefore, the findings of the study may not be generalisable to all pregnant women.

In spite of these limitations, these findings encourage dietary counselling at the antenatal visits for all pregnant women with elevated risk, as it could influence their dietary intakes for better and potentially that of their family members as well. Usually, mothers-to-be are keen on making lifestyle changes that could benefit their children. However, it is evident that low-intensity counselling leads to only modest changes; this phenomenon has also been seen in T2D prevention studies.[29]

Larger randomised studies are required to fully understand how dietary counselling should be implemented and targeted. For example, it may be worth exploring whether individual online sessions with a dietitian would be more efficient than visiting the dietitian in person, especially among women with small children. In addition, studies such as Guelinckx,[10] exploring the degree of intervention needed to create dietary changes, could help decide how much counselling should be given. The same counselling method may not work for all pregnant women; hence, examining the influence of various counselling methods in various subgroups of women could help in individualising the counselling sessions. Finally, exploring the effect of dietary counselling initiated already before pregnancy is also relevant in future studies.

In conclusion, the findings of this study showed that dietary counselling in early pregnancy, in addition to standard antenatal care, can lead to modest dietary improvements among women at GDM risk.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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AUTHOR CONTRIBUTIONS

AV, SK, JM, JL, BS-L, KR, MK, HK, MP-A, AT, EH, HL, SA, RR and JE designed and conducted the study; AV and HK analysed the data; AV interpreted the data and wrote the manuscript; SK, JM, JL, BS-L, KR, MK, HK, MP-A, AT, EH, HL, SA, RR and JE gave their comments of the first version of the manuscript. All authors approved the final version of the manuscript to be published.

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